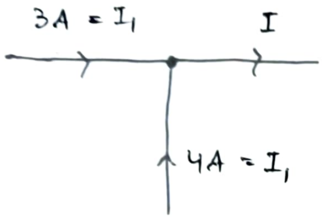
VECTOR ALGEBRA

* Physical Quantities :-

→ **Scalar Quantities**
- → only magnitude
- → No sense of orientation or Not associated with dir^n

Eg:- Potential (V), Energy (E), Work (W), etc.

→ **Vector Quantities**
- → magnitude
- → dir^n associated with it
- → Also follow Vector Law of Addition



$I = I_1 + I_2 = 3A + 4A$ ⟹ Algebric Law of Add^n

$\therefore$ I is not a vector quantity.

Eg:- Force ($\vec{F}$), Electric Field ($\vec{E}$), Magnetic Field. ($\vec{H}$)

→ **Tensor Quantity**
- → magnitude
- → associated with dir^n
- → Follow algebric Law of addition

Eg:- Current (I), Pressure (P), etc.

VECTORS

Representation :- $\boxed{\vec{A} = |\vec{A}| \cdot \hat{a}_A}$ → Direction

$|\vec{A}|$ → Magnitude

$$\boxed{\hat{a}_A = \frac{\vec{A}}{|\vec{A}|}}$$

unit vector of main vector = dir^n of main vector

Direction ⟷ Unit Vectors **

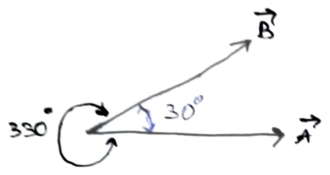
Unit vector :- It is also dirⁿ in which main vector increases.

$\hat{a}_x$, $\hat{a}_y$, $\hat{a}_z$ are unit vetor in x, y, z - axis.

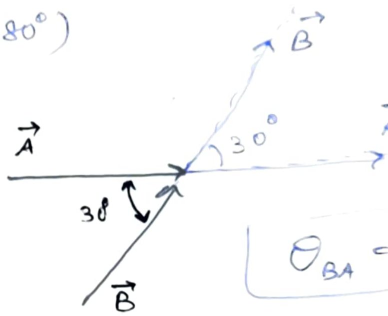
Angle between two vectors

1) Tail - Tail joined vectors

2) $0 \leq \theta_A \leq 180^\circ$ $(\theta_A \ngtr 180^\circ)$



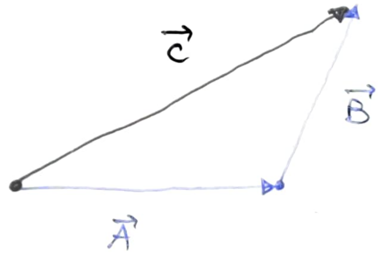
$\boxed{\theta_{AB} = 30^\circ}$



$\boxed{\theta_{BA} = \theta_{AB} = 30^\circ}$

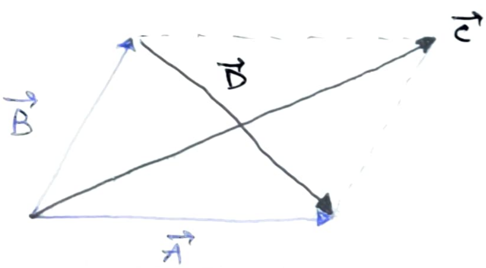
VECTOR LAW OF ADDITION

1) Trianglur Law of Addⁿ



$\boxed{\vec{C} = \vec{A} + \vec{B}}$

2) Parallelogram Law

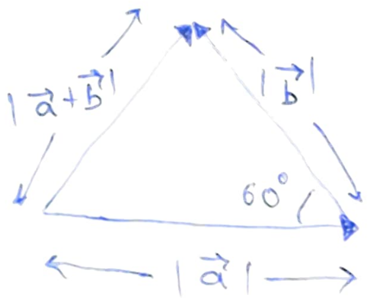


$\boxed{\vec{C} = \vec{A} + \vec{B}}$

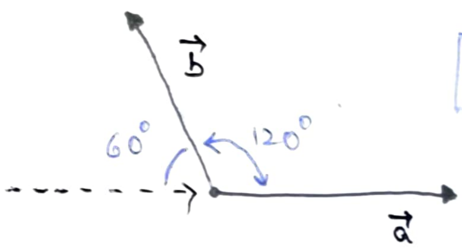
* $\boxed{\vec{D} = \vec{A} - \vec{B}}$

Q :- $\vec{a}$ & $\vec{b}$ are unit vectors and α is the angle b/w them, Calculate value of α such that $\vec{a} + \vec{b}$ is also unit vector.

Sol :- $|\vec{a}| = 1$, $|\vec{b}| = 1$, $|\vec{a} + \vec{b}| = 1$



$\therefore \theta = 60^\circ$ but α is



$\boxed{\alpha = 120^\circ}$

It is an equilateral Δ.

MULTIPLICATION OF VECTORS

- Dot Product (Scalar Product)
- Cross Product (Vector Product)

1) Dot Products

$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta_{AB}$ $\qquad 0 \le \theta_{AB} \le 180^\circ$

$\vec{B} \cdot \vec{A} = |\vec{B}||\vec{A}| \cos\theta_{BA}$ $\qquad \boxed{\theta_{AB} = \theta_{BA}}$

** $\boxed{\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}}$

Properties :-

* $\vec{A} \cdot \vec{A} = |\vec{A}|^2 = A^2$
* $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
* $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$
* $\vec{A} \cdot \vec{B} = AB\cos\theta$

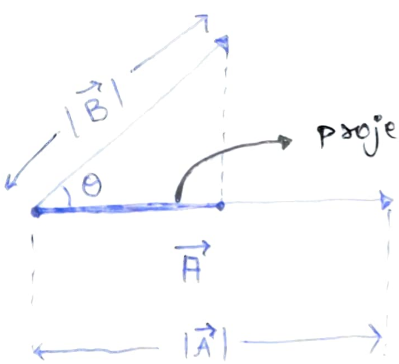
→ If $\vec{A} \perp \vec{B} \Rightarrow \theta = 90^\circ \Rightarrow \vec{A} \cdot \vec{B} = 0$ (Orthogonal)

→ If $\vec{A} \parallel \vec{B} \Rightarrow \theta = 0^\circ \Rightarrow \vec{A} \cdot \vec{B} = AB$ (Parallel)

→ If $\vec{A} \uparrow\downarrow \vec{B} \Rightarrow \theta = 180^\circ \Rightarrow \vec{A} \cdot \vec{B} = -AB$ (Anti-Parallel)

• Geometrical Significance of Dot Product

Concept of Projection :-



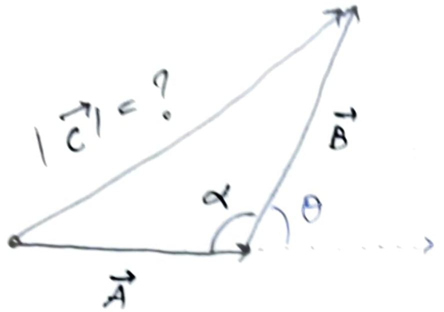
projection of $\vec{B}$ on $\vec{A}$ = component of $\vec{B}$ in dir^n of $\vec{A}$

$= |\vec{B}| \cos\theta$

$= B\cos\theta$

$$\vec{A} \cdot \vec{B} = \underbrace{|\vec{A}|}_{\text{Length of } \vec{A}} \overbrace{(|\vec{B}|\cos\theta)}^{\text{Projection of } \vec{B} \text{ on } \vec{A}}$$

Q :-



Find $|\vec{C}|$ in terms of α ?

as, $\theta + \alpha = \pi$

$[\alpha = \pi - \theta]$

Also,
$$\vec{C} \cdot \vec{C} = |\vec{C}|^2 = (\vec{A} + \vec{B}) \cdot (\vec{A} + \vec{B})$$
$$= |\vec{A}|^2 + |\vec{B}|^2 + \vec{B} \cdot \vec{A} + \vec{A} \cdot \vec{B}$$
$$C^2 = A^2 + B^2 + 2AB\cos\theta$$
$$C^2 = A^2 + B^2 + 2AB\cos(\pi - \alpha)$$
$$\boxed{C^2 = A^2 + B^2 - 2AB\cos\alpha}$$

* **Note** :- Here, θ is the angle b/w $\vec{A}$ & $\vec{B}$

α is the given angle

Be careful which angle is used

Q :- i) $\vec{a} + \vec{b} + \vec{c} = 0$, Find $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = ?$

$$(\vec{a} + \vec{b} + \vec{c})^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a}$$
$$0 = a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$
$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\left[\frac{|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2}{2}\right]$$

ii) If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

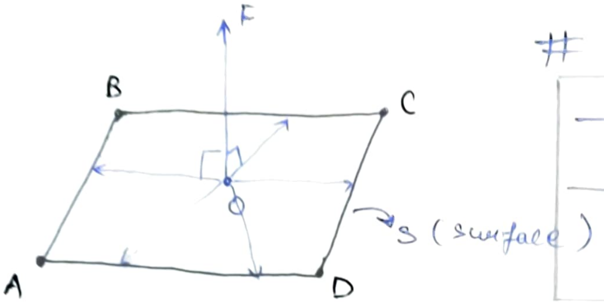
H.W :- Q :-

$\cos\frac{\theta}{2} = \frac{1}{2}|\vec{a} + \vec{b}| \rightarrow$ Prove this

$\sin\frac{\theta}{2} = \frac{1}{2}|\vec{a} - \vec{b}| \rightarrow$ Prove this

$\vec{a}$ & $\vec{b}$ are unit vector & θ is angle b/w them.

Imp. point (useful in "spherical coordinate")



Surface ABCD

→ $\overrightarrow{OF} \perp$ surface ABCD

→ $\overrightarrow{OF} \perp$ (Any vector drawn on surface ABCD)

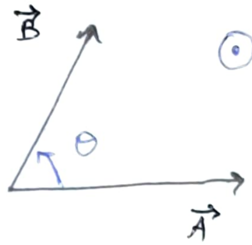
2) CROSS PRODUCT

$$\boxed{\vec{A} \times \vec{B} = \underbrace{|\vec{A}|\,|\vec{B}|\sin\theta}_{\text{Magnitude}}\ \underbrace{\hat{a}_n}_{\text{direction}} = AB\sin\theta\ \hat{a}_n}$$

i) $|\vec{C}| = |\vec{A} \times \vec{B}| = AB\sin\theta$

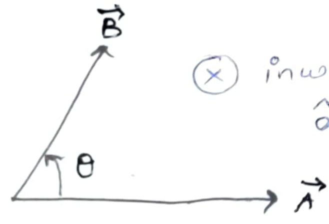
ii) $\hat{a}_c = \hat{a}_{A\times B} = \hat{a}_n \Rightarrow$ Unit vector $\perp$ to both $\vec{A}$ & $\vec{B}$

iii) Right Handed System $\Rightarrow$ Right Hand Thumb rule



$\odot$ outside screen $\hat{a}_n$

$(\vec{A} \times \vec{B})$



$\otimes$ inward screen $\hat{a}_n$

- Properties

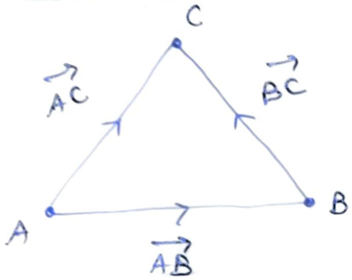
* $(\vec{A} \times \vec{B}) \parallel \hat{a}_n$
* $(\vec{A} \times \vec{B}) \perp \vec{A}$
* $(\vec{A} \times \vec{B}) \perp \vec{B}$
* $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$
* $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$
* If $\vec{A} \parallel \vec{B} \Rightarrow \boxed{\theta_{AB} = 0^\circ \Rightarrow \vec{A} \times \vec{B} = 0}$

  (Also in case of anti-parallel $\theta = 180^\circ$)
* $\vec{A} \perp \vec{B} \Rightarrow \theta_{AB} = 90^\circ \Rightarrow \boxed{\vec{A} \times \vec{B} = AB\ \hat{a}_n}$

• Physical significance of Vector product

1) Area of triangle :-

$$\boxed{\frac{1}{2}\left|\vec{AB} \times \vec{AC}\right|}$$

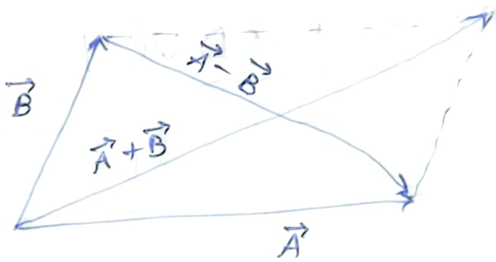


2) Area of parallelogram :-

Area $= |\vec{A} \times \vec{B}|$

Area $= \frac{1}{2}\left|(\vec{A}+\vec{B}) \times (\vec{A}-\vec{B})\right|$

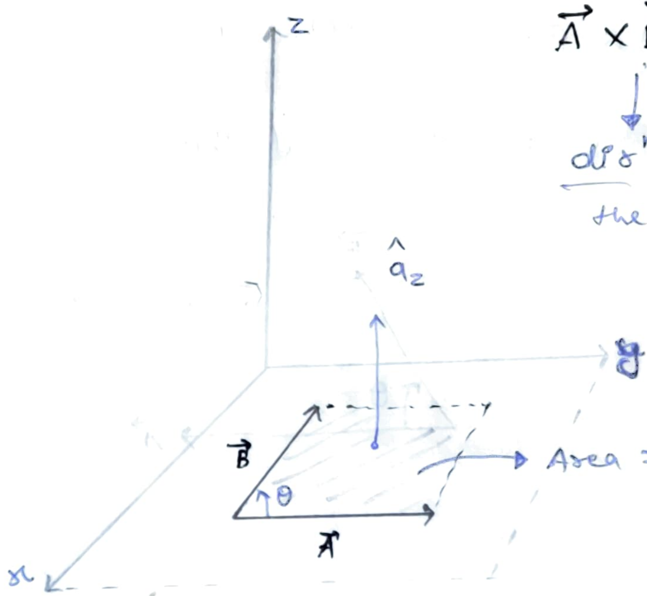
→ with help of diagonal



3) Concept of Area Vector :

$\vec{A} \times \vec{B}$ → Mag. = Area of $\parallel^{gm}$ formed in plane of $\vec{A}$ & $\vec{B}$

dir^n = $\perp$ to the area following RHS



→ Area = $AB\sin\theta$

★ Imp. point

$\vec{A}\cdot\vec{B} = AB\cos\theta \rightarrow \theta = \cos^{-1}\left[\frac{\vec{A}\cdot\vec{B}}{AB}\right]$

$\vec{A}\times\vec{B} = AB\sin\theta \rightarrow \boxed{\theta = \sin^{-1}\left[\frac{\vec{A}\times\vec{B}}{AB}\right]}$

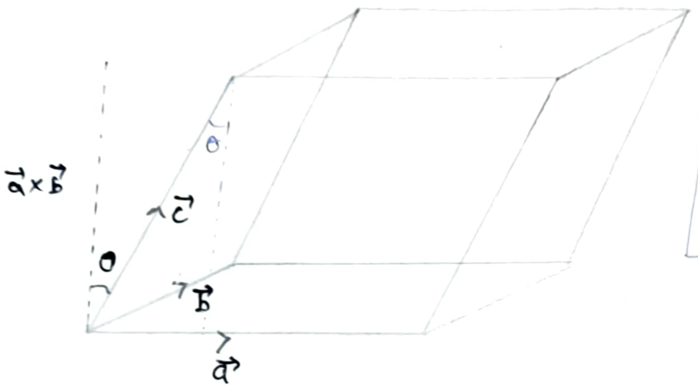
→ More appropriate formula

→ Mostly use this

3> SCALAR TRIPLE PRODUCT (STP)

$$[\vec{A}\ \vec{B}\ \vec{C}] = \vec{A}\cdot(\vec{B}\times\vec{C}) = \vec{B}\cdot(\vec{C}\times\vec{A}) = \vec{C}\cdot(\vec{A}\times\vec{B})$$

$$\vec{A}\cdot(\vec{B}\times\vec{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

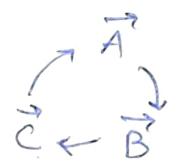


$$\boxed{\text{Volume of Parallelopiped} = \vec{A}\cdot(\vec{B}\times\vec{C})}$$

- **Properties** :-

* cyclic permute of STP is same

$$[\vec{A}\ \vec{B}\ \vec{C}] = [\vec{B}\ \vec{C}\ \vec{A}] = [\vec{C}\ \vec{A}\ \vec{B}]$$



* $[\vec{a}\ \vec{b}\ \vec{c}] = 0$ (Non-zero, non-collinear).

4> VECTOR TRIPLE PRODUCT

$$\boxed{\vec{A}\times(\vec{B}\times\vec{C}) = (\vec{A}\cdot\vec{C})\vec{B} - (\vec{A}\cdot\vec{B})\vec{C}}$$

But use, $\vec{B}\times\vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \end{vmatrix} = \vec{Z}$

$\vec{A}\times\vec{Z} = \begin{vmatrix} & & \\ & & \\ & & \end{vmatrix} =$ Final

COORDINATE SYSTEM (3-D)

1) Rectangular or Cartesian CS
2) Cylindrical CS
3) Spherical CS

$\Longrightarrow$ used to locate a point in 3-D

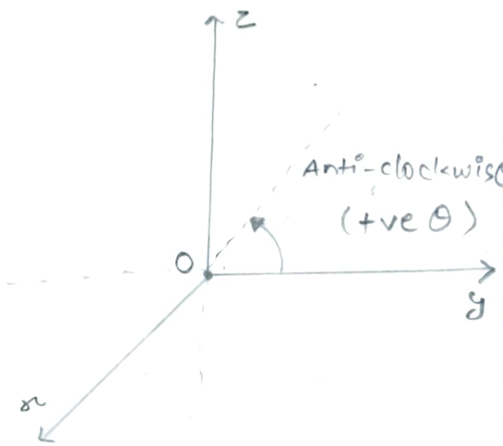
RECTANGULAR COORDINATE SYSTEM

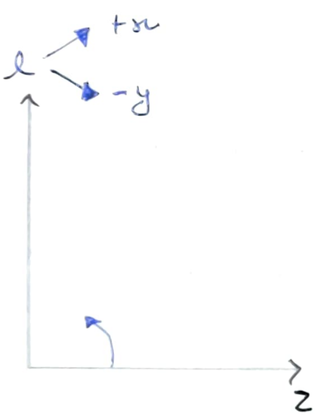
* Parameter :- x, y, z

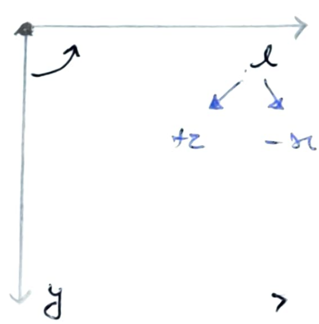
$-\infty < x < \infty$

$-\infty < y < \infty$

$-\infty < z < \infty$







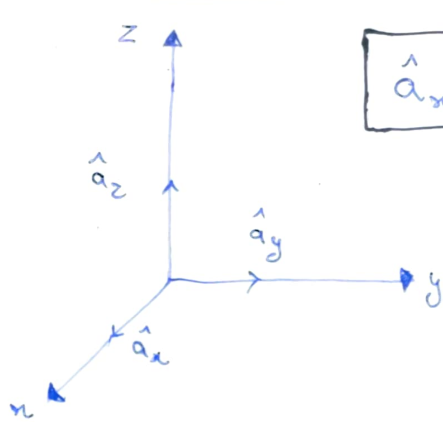
★ R.C.S is Right Handed System.

* Basis Vector or Set of unit vector along parameter of a coordinate system

P(x, y, z)

$\downarrow$

along which parameter ↑es



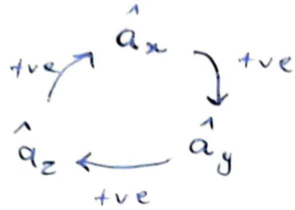
$$\boxed{\hat{a}_x \perp \hat{a}_y \perp \hat{a}_z}$$

3> Dot Product and cross Product of unit vectors

\# $\boxed{\hat{a}_x \cdot \hat{a}_y = \hat{a}_y \cdot \hat{a}_z = \hat{a}_z \cdot \hat{a}_x = 0}$

\# $\boxed{\hat{a}_x \cdot \hat{a}_x = \hat{a}_y \cdot \hat{a}_y = \hat{a}_z \cdot \hat{a}_z = 1}$

$$\boxed{\begin{aligned} \hat{a}_x \times \hat{a}_x &= 0 \\ \hat{a}_y \times \hat{a}_y &= 0 \\ \hat{a}_z \times \hat{a}_z &= 0 \end{aligned}}$$



$$\hat{a}_x \times \hat{a}_y = \hat{a}_z$$
$$\hat{a}_y \times \hat{a}_z = \hat{a}_x$$
$$\hat{a}_z \times \hat{a}_x = \hat{a}_y$$

4> Let $\vec{A} = \{\vec{A}_1, \vec{A}_2, \ldots \vec{A}_m\}$ → set of m vectors

If $\begin{cases} A_i \cdot A_j \neq 0, & i = j \\ A_i \cdot A_j = 0, & i \neq j \end{cases}$ ⇒ $\vec{A}$ represent set of orthogonal vectors

If $\begin{cases} A_i \cdot A_j = 1, & i = j \\ A_i \cdot A_j = 0, & i \neq j \end{cases}$ ⇒ $\vec{A}$ represent set of orthonormal vectors

* Vector Representation in Rect. CS :-

$$\boxed{\vec{A} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z}$$

component of $\vec{A}$ in x, y, z dir^n respectively

$$\boxed{|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}}$$ → Magnitude

$$\boxed{\hat{a}_A = \frac{A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}}$$ → direction

$f(x) = \sin x \rightarrow$ Scalar Function

$f(x) = (\sin x)\hat{a}_x + (\cos x)\hat{a}_y \rightarrow$ vector Function

$$\boxed{\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z}$$

$\nearrow$ $x, y, z \rightarrow$ variable : "position vector Function"

$\searrow$ $x, y, z \rightarrow$ constant : position vector

Locus In Cartesian System

Case 1 :

$$\left.\begin{aligned} x &= a \\ y &= b \\ z &= c \end{aligned}\right\} \text{all constant}$$

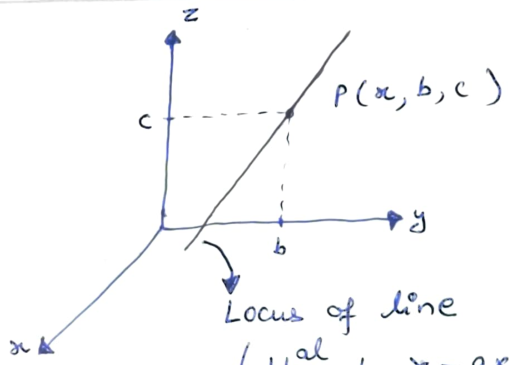
$P(a, b, c)$

$\downarrow$

Locus of a point

Case 2 :

$$\left.\begin{aligned} x &= \text{variable} \\ y &= b \\ z &= c \end{aligned}\right\}$$



Locus of line ($\|^{al}$ to x-axis)

$$\left.\begin{aligned} x &= a \\ y &= \text{variable} \\ z &= c \end{aligned}\right\} \text{Line } \|^{al} \text{ to } y\text{-axis}$$

$$\left.\begin{aligned} x &= a \\ y &= b \\ z &= \text{variable} \end{aligned}\right\} \text{Line } \|^{al} \text{ to } z\text{-axis}$$

Case 3 :

$$\left.\begin{aligned} x &= a \\ y &= \text{variable} \\ z &= \text{variable} \end{aligned}\right\}$$

$P(x, y, z) = P(a, y, z)$

$\hookrightarrow$ surface / Plane / Area surface

Surface $\Rightarrow$ 2-D curve ; 2 parameter of certain C.S changes

$\hookrightarrow$ a vector quantity

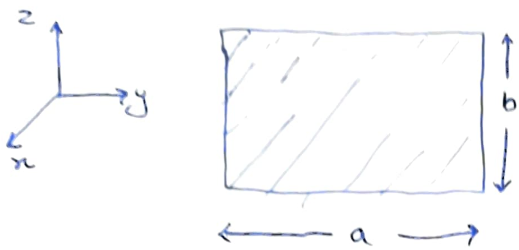
$$\boxed{\vec{S} = |\vec{S}|\,\hat{a}_s}$$

$|\vec{S}|$ = Area of surface

$\hat{a}_s$ = direction of surface

How to calculate $\hat{a}_s$?

Case 1 : Let "S" is open surface without orientation.

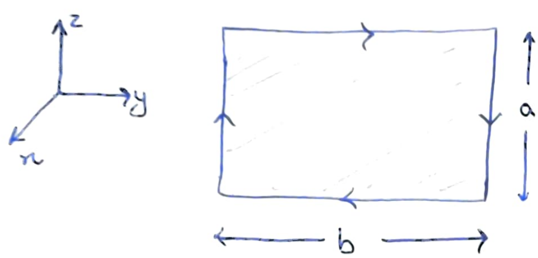


$$\vec{S} = |\vec{S}|\,\hat{a}_s \overset{**}{=} |\vec{S}|(\pm\,\hat{a}_x)$$

i) $|\vec{S}| = ab$

ii) $\hat{a}_s$ = perpendicular to the surface

Case 2 : If "S" is open surface with orientation.



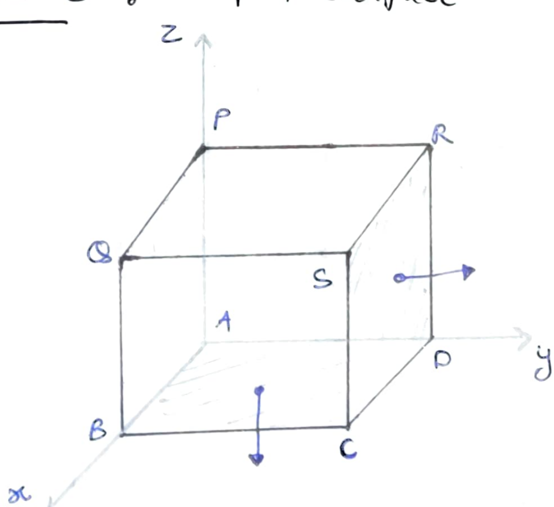
$$\vec{S} = |\vec{S}|\,\hat{a}_s = |\vec{S}|(-\hat{a}_x)$$

$|\vec{S}| = ab$

$\hat{a}_s$ = perpendicular to surface & following ~~RHGH~~ $\boxed{\text{RIGHT HANDED SYSTEM}}$

$$\boxed{\vec{S} = (ab)(-\hat{a}_x)}$$

Case 3 : "Open surface" which is part of volume :



Surface : # CDRS

$$\vec{S} = |\vec{S}|\,\hat{a}_s = |\vec{S}|(+\hat{a}_y)$$

$|\vec{S}| = \checkmark$

$\hat{a}_s$ = perpendicular such that it is pointing

** $\boxed{\text{outward volume.}}$

ABCD : $\vec{S} = |\vec{S}|(-\hat{a}_z)$

$\Rightarrow$ $x = a$

$y \rightarrow$ variable

$z \rightarrow$ variable

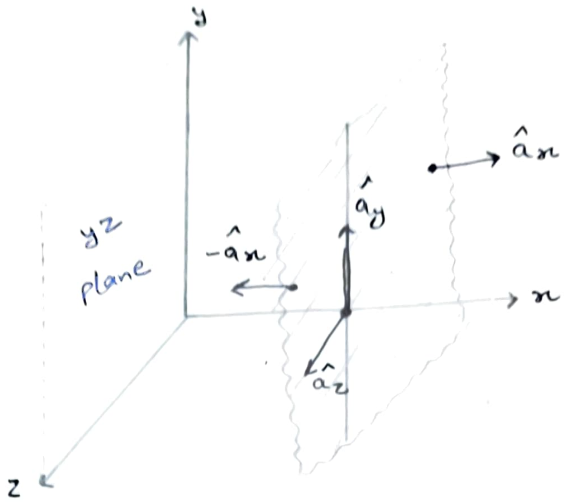
$\}$ $\rightarrow$ plane $\quad P(x, y, z) = P(a, y, z)$

$\Downarrow$

$x = a$

$\downarrow$

eq^n of a surface parallel to yz plane



Summary

1. $\boxed{x = a}$ $\Rightarrow$ a plane parallel to yz plane.

   $\Rightarrow$ unit vector $\perp^r$ to $x = a$ $= (\pm \hat{a}_x)$

   $\Rightarrow$ unit vector $\parallel^{al}$ to $x = a$ $= (\pm \hat{a}_y, \pm \hat{a}_z)$

2. $\boxed{y = b}$ $\Rightarrow$ a plane parallel to xz plane

   $\Rightarrow$ unit vector $\perp^r$ to $y = b$ $(\pm \hat{a}_y)$

   $\Rightarrow$ unit vector $\parallel^{al}$ to $y = b$ $(\pm \hat{a}_x, \pm \hat{a}_z)$

3. $\boxed{z = c}$ $\Rightarrow$ a plane $\parallel^{al}$ to xy-plane

   $\Rightarrow$ unit vector $\perp^r$ to $z = c$ $(\pm \hat{a}_z)$

   $\Rightarrow$ unit vector $\parallel^{al}$ to $z = c$ $(\pm \hat{a}_x, \pm \hat{a}_y)$

**

Important

$P(l, m, n) \rightarrow$ In any coordinate system

$l = c \rightarrow$ plane

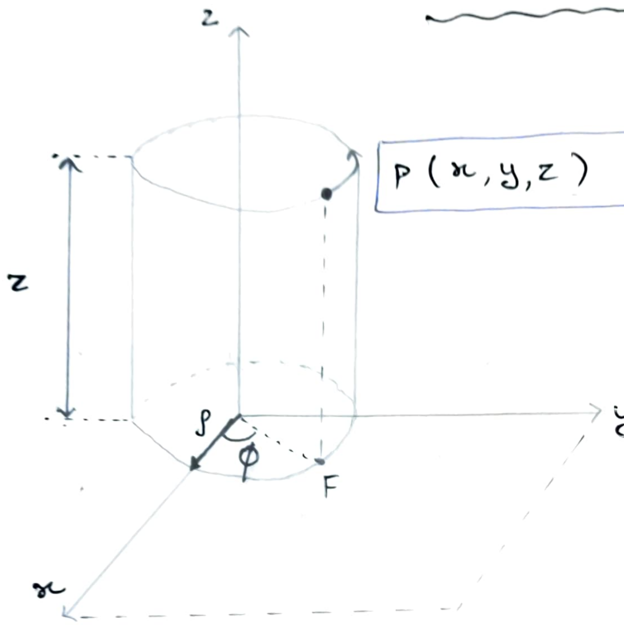
$\rightarrow$ unit vector $\perp^r$ to $l = c$ $\Rightarrow (\pm \hat{a}_l)$

Ques # spherical CS : $P(r, \theta, \phi)$

$\Rightarrow$ unit vector $\perp^r$ to $\theta = C$ surface $\rightarrow$ $(\pm \hat{a}_\theta)$

$\Rightarrow$ direction of $\phi = C$ surface $\rightarrow$ $(\pm \hat{a}_\phi)$

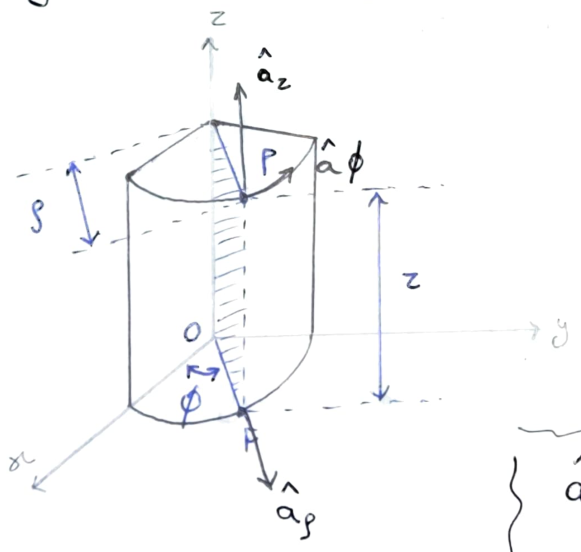
"CYLINDRICAL COORDINATE SYSTEM"



$$\boxed{P(x, y, z) = P(\rho, \phi, z)}$$

F = Foot of $\perp^r$ of point P in xy-plane

1) Any point in cylindrical CS is rep. by $P(\rho, \phi, z)$



$$\boxed{P(\rho, \phi, z)}$$

$$\boxed{\hat{a}_\rho \perp \hat{a}_\phi \perp \hat{a}_z}$$

$\hat{a}_\rho \rightarrow$ radial dir^n

$\hat{a}_\phi \rightarrow$ tangential dir^n

$\rho \rightarrow$ radius of imaginary cylinder passing through point (P) or its foot of $\perp^r$ (F).

$$\boxed{0 \leq \rho < \infty}$$

$\phi \rightarrow$ Azimuthal angle : "measured from +ve x axis towards vector $\vec{OF}$ in ACW dir^n."

$$\boxed{0 \leq \phi < 2\pi}$$

$z \rightarrow$ same as cartesian / rectangular C.S

$$\boxed{-\infty < z < -\infty}$$

2) Basis Vector or Unit vector in cylindrical C.S

$\hat{a}_\rho \rightarrow$ unit vector in dir^n of ρ increases

$\hat{a}_\phi \rightarrow$ " " " " " ϕ "

$\hat{a}_z \rightarrow$ " " " " " z "

3) Dot product & Cross product

Dot product $\Rightarrow$ Diff = 0

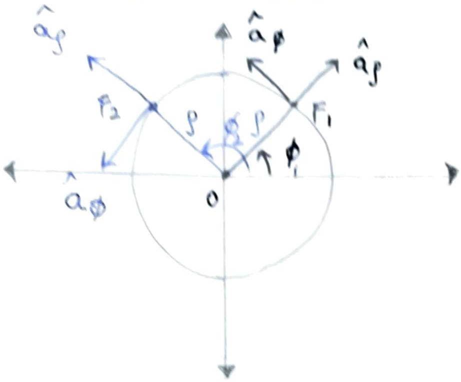
same = 1

Cross product $\Rightarrow$ same = 0

diff = 3rd following RHS

$$\begin{tikzcd} & a_\rho \arrow[dr] & \\ a_z \arrow[ur] & & a_\phi \arrow[ll] \end{tikzcd}$$

4) Locus of Foot of $\perp^r$ of points $\Rightarrow$ xy plane



at F_1 : $\vec{\rho_1} = \rho \cdot \hat{a}_{\rho_1}$

$\vec{\phi_1} = \phi_1 \, \hat{a}_{\phi_1}$

at F_2 : $\vec{\rho_2} = \rho \cdot \hat{a}_{\rho_2}$

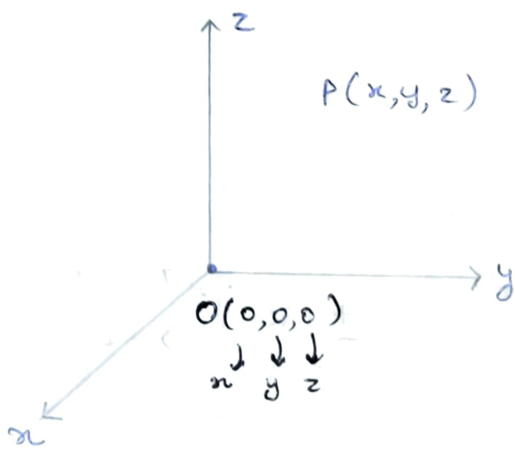
$\vec{\phi_2} = \phi_2 \cdot \hat{a}_{\phi_2}$

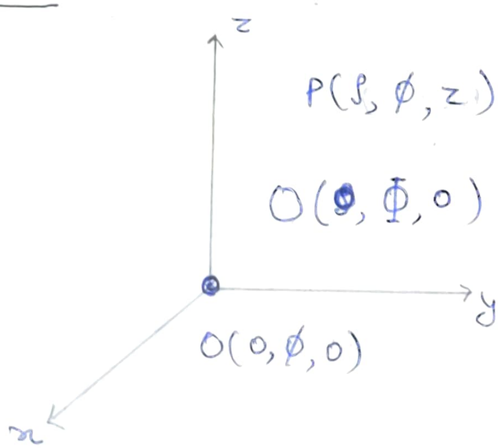
Note : $P(\rho, \phi, z)$

as ϕ changes → $\hat{a}_\rho$ changes
→ $\hat{a}_\phi$ changes
→ ρ does not changes

$$\boxed{\begin{aligned} \hat{a}_\rho &= f(\phi) \\ \hat{a}_\phi &= f(\phi) \end{aligned}}$$

COORDINATES OF ORIGIN





Vector Representation in Cylindrical C.S

$$\boxed{\vec{A} = A_\rho \, \hat{a}_\rho + A_\phi \, \hat{a}_\phi + A_z \, \hat{a}_z}$$

Component of unit vector along $\hat{a}_\rho$, $\hat{a}_\phi$, $\hat{a}_z$ respectively

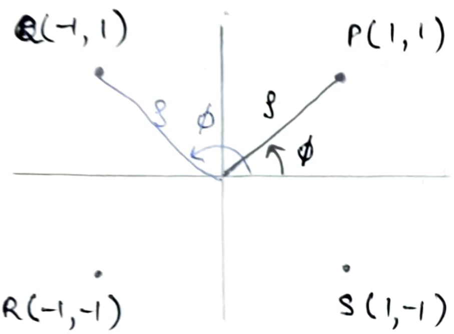
Conversion Of Cylindrical C.S to Cartesian C.S

1) Point conversion
2) Vector conversion
3) Unit vector conversion

1) POINT CONVERSION

$$\boxed{\begin{aligned} x &= \rho\cos\phi \\ y &= \rho\sin\phi \\ z &= z \end{aligned}}$$ → From Cylindrical to cartesian

$$\boxed{\begin{aligned} \rho &= \sqrt{x^2+y^2} \\ \phi &= \tan^{-1}(y/x) \\ z &= z \end{aligned}}$$ → From cartesian to cylindrical



I ⇒ at P : $\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{1}\right) = 45°$

II ⇒ at Q : $\phi = \tan^{-1}\left(\frac{1}{-1}\right) = -45°$ $\pm 180°$ → $135°$ → $-225°$

III ⇒ at R : $\phi = \tan^{-1}\left(\frac{-1}{-1}\right) = 45$ $\pm 180°$ → $225°$ → $-135°$

IV ⇒ at S : $\phi = \tan^{-1}\left(\frac{-1}{1}\right) = -45°$ (No encirculement)

Ques :- $P(-1, 1, \sqrt{2})$ ⇒ Cartesian CoS to Cylindrical

$x = -1$ ⇒ II^{nd} quad.

$y = 1$

$z = \sqrt{2}$

$\rho = \sqrt{(-1)^2+1^2} = \sqrt{2}$

$\phi = \tan^{-1}\left(\frac{1}{-1}\right) = -45° + 180°$

$= 135°$

$z = \sqrt{2}$

** $\boxed{P(-1, 1, \sqrt{2}) \equiv P(\sqrt{2}, 135°, \sqrt{2})}$

Vector Conversion

$$\vec{A}_{cart} = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z$$

$$A_{cylind.} = A_\rho \hat{a}_\rho + A_\phi \hat{a}_\phi + A_z \hat{a}_z$$

Inter-conversion

$$\boxed{\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}}$$

$$\text{Det} = 1 \quad , \quad A^{-1} = A^T \text{ (Transpose)}$$

$$\boxed{\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}}$$

$$\vec{A}_{cart} = x \hat{a}_x + y \hat{a}_y + z \hat{a}_z \rightarrow \text{Cartesian}$$

$$\vec{A}_{cylin} = A_\rho \hat{a}_\rho + A_\phi \hat{a}_\phi + A_z \hat{a}_z \rightarrow \text{cylindrical}$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{aligned} x &= \rho\cos\phi \\ y &= \rho\sin\phi \\ z &= z \end{aligned}$$

$$\therefore \quad A_\rho = x\cos\phi + y\sin\phi + 0(z)$$

$$A_\rho = \rho\cos^2\phi + \rho\sin^2\phi = \boxed{\rho}$$

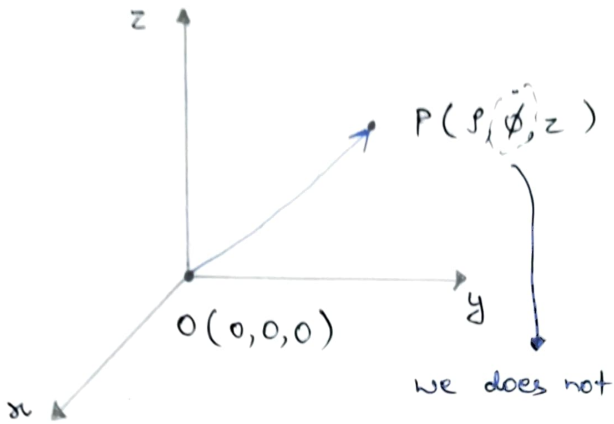
Also, $A_\phi = -x\sin\phi + y\cos\phi + 0 = -\rho\cos\phi\sin\phi + \rho\cos\phi \ldots$

$$\boxed{A_\phi = 0}$$

And, $\boxed{A_z = z}$

$$\vec{A}_{cart} = x\,\hat{a}_x + y\,\hat{a}_y + z\,\hat{a}_z \equiv \vec{A}_{cylin} = \rho\,\hat{a}_\rho + 0\,\hat{a}_\phi + z\,\hat{a}_z$$

* Position Vector in Cylindrical C.S



$$\vec{r}_p = (\rho - 0)\,\hat{a}_\rho + (z-0)\,\hat{a}_z$$

$$\boxed{\vec{r}_p = \rho\,\hat{a}_\rho + z\,\hat{a}_z}$$

we does not consider ϕ, in position vector calculation.

* Resultant Vector in Cylindrical C.S :

$1(\rho_1, \phi_1, z_1)$ $\qquad$ $2(\rho_2, \phi_2, z_2)$

$$\boxed{\vec{r}_{12} = (\rho_2 - \rho_1)\,\hat{a}_\rho + (z_2 - z_1)\,\hat{a}_z}$$

UNIT VECTOR CONVERSION

$x = \rho\cos\phi$

$y = \rho\sin\phi$

$z = z$

$$\begin{bmatrix} \hat{a}_x \\ \hat{a}_y \\ \hat{a}_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{a}_\rho \\ \hat{a}_\phi \\ \hat{a}_z \end{bmatrix}$$

**

$$\hat{a}_\rho = \cos\phi\,\hat{a}_x + \sin\phi\,\hat{a}_y = \boxed{f(\phi)}$$

$$\hat{a}_\phi = -\sin\phi\,\hat{a}_x + \cos\phi\,\hat{a}_y = \boxed{f(\phi)}$$

$$\begin{bmatrix} \hat{a}_\rho \\ \hat{a}_\phi \\ \hat{a}_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{a}_x \\ \hat{a}_y \\ \hat{a}_z \end{bmatrix}$$

$$\hat{a}_z = \hat{a}_z$$

Ques : $\int_0^{2\pi} \hat{a}_\rho \, d\phi = ?$

$\hat{a}_\rho \int_0^{2\pi} d\phi = 2\pi \hat{a}_\rho$ (Wrong) ✗

Solⁿ :-

$$\int_0^{2\pi} (\cos\phi \, \hat{a}_x + \sin\phi \, \hat{a}_y) \, d\phi$$

$$\hat{a}_x \underbrace{\int_0^{2\pi} \cos\phi \, d\phi}_{0} + \hat{a}_y \underbrace{\int_0^{2\pi} \sin\phi \, d\phi}_{0} = 0$$

* Area under the curve $\sin\theta$ & $\cos\theta$ in $[0, 2\pi]$ is 0.

4) Dot Product of unit vectors in diff. C.S

$x = \rho \cos\phi$

$y = \rho \sin\phi$

$z = z$

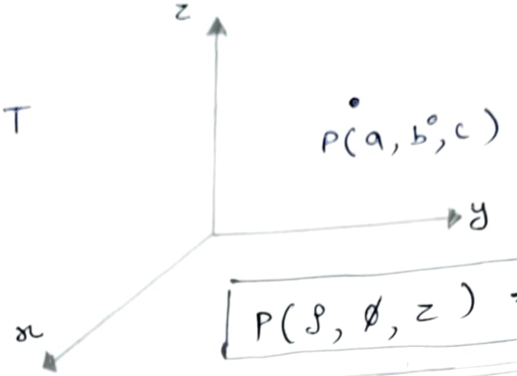
	$\hat{a}_\rho$	$\hat{a}_\phi$	$\hat{a}_z$
$\hat{a}_x$	$\cos\phi$	$-\sin\phi$	0
$\hat{a}_y$	$\sin\phi$	$\cos\phi$	0
$\hat{a}_z$	0	0	1

* $\hat{a}_x \cdot \hat{a}_\phi = -\sin\phi$

* $\hat{a}_y \cdot \hat{a}_\rho = \sin\phi$

LOCUS in CYLINDRICAL S/S :-

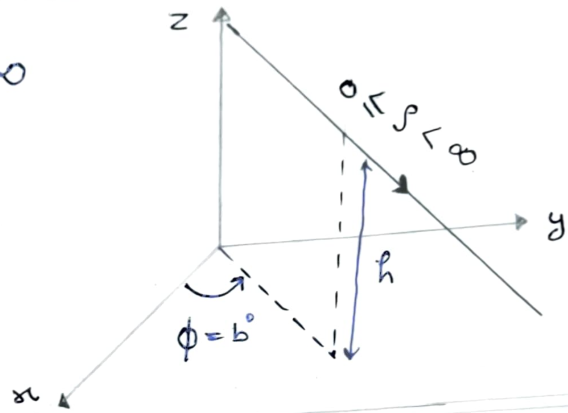
1) $\rho = a$ → POINT
$\phi = b^\circ$
$z = c$



$\boxed{P(\rho, \phi, z) = P(a, b^\circ, c)}$

2) $\rho \to$ changing $\Rightarrow 0 \leq \rho < \infty$

$\phi = b^\circ$
$z = c$ (RADIAL LINE)



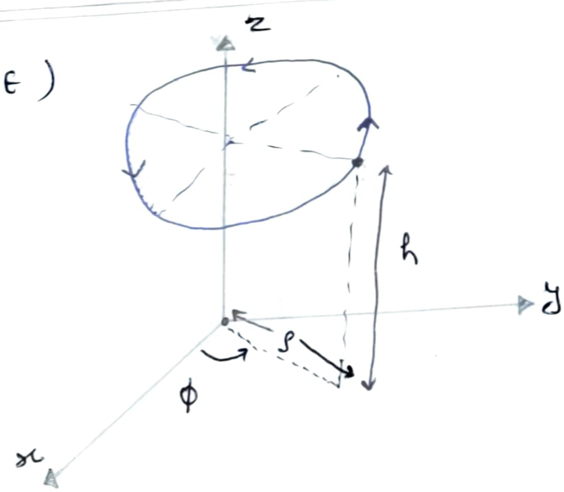
3) $\rho \to a$ (CIRCULAR LINE)

$\phi \to$ variable $0 \leq \phi < 2\pi$

$z \to$ const.

→ circle of radius a

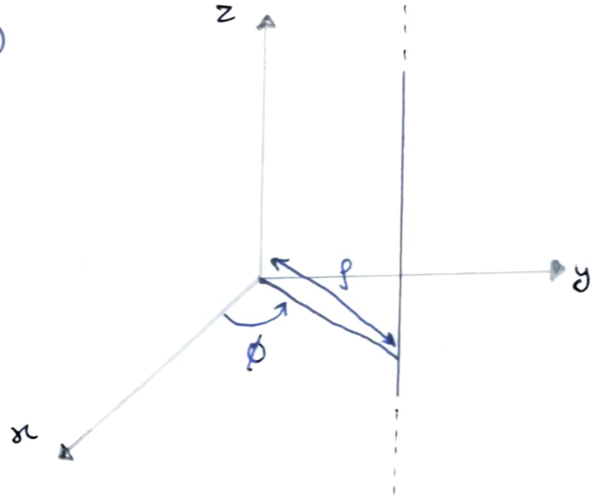
→ it will be present in $z = h$ plane



4) $\rho = a$ (STRIGHT LINE)

$\phi = b^\circ$

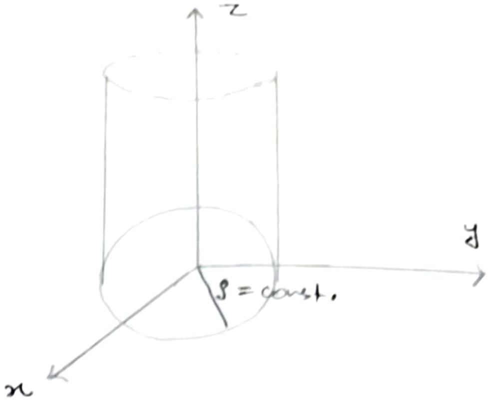
$z =$ variable



5) $\boxed{\begin{array}{l} \rho = \text{constant} \\ \phi = \text{variable} \\ z = \text{variable} \end{array}}$ $\begin{array}{l} \\ \rightarrow 0 \le \phi < 2\pi \\ \rightarrow -\infty < z < \infty \end{array}$

$\Downarrow$

Hollow Cylinder



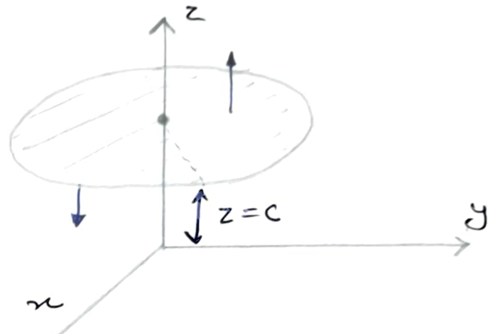
Unit vector $\perp^r$ to $\rho = c \Rightarrow \hat{a}_\rho$

---

6) $\boxed{\begin{array}{l} \rho \rightarrow \text{variable} \\ \phi \rightarrow \text{variable} \\ z \rightarrow \text{const.} \end{array}}$ $\begin{array}{l} \rightarrow 0 \le \rho < \infty \\ \rightarrow 0 \le \phi < 2\pi \\ \\ \end{array}$

$\Downarrow$

Infinite Disk



Unit Vector $\perp^r$ to surface or Dir^n $\Rightarrow \pm \hat{a}_z$

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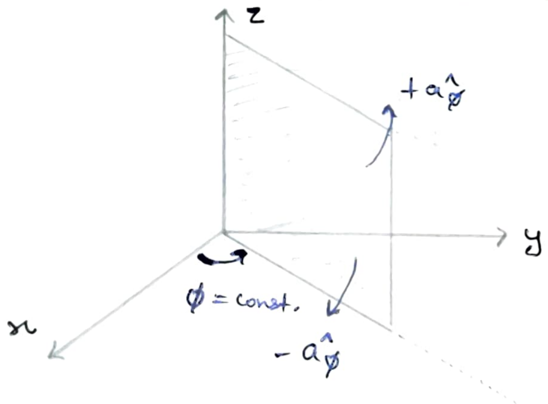
7) $\boxed{\begin{array}{l} \rho \rightarrow \text{variable} \\ \phi \rightarrow \text{const.} \\ z \rightarrow \text{variable} \end{array}}$



Unit Vector $\perp^r$ to surface $\Rightarrow \pm \hat{a}_\phi$

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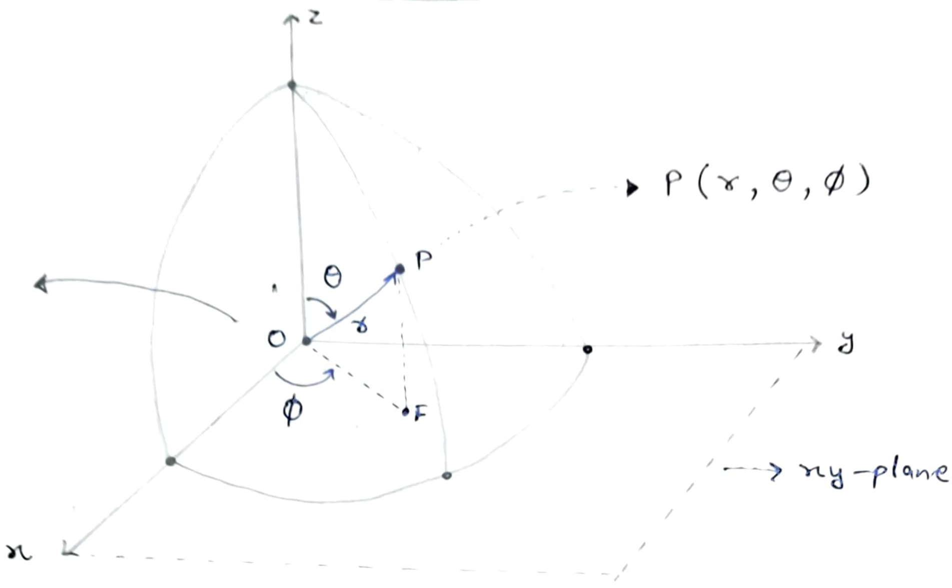
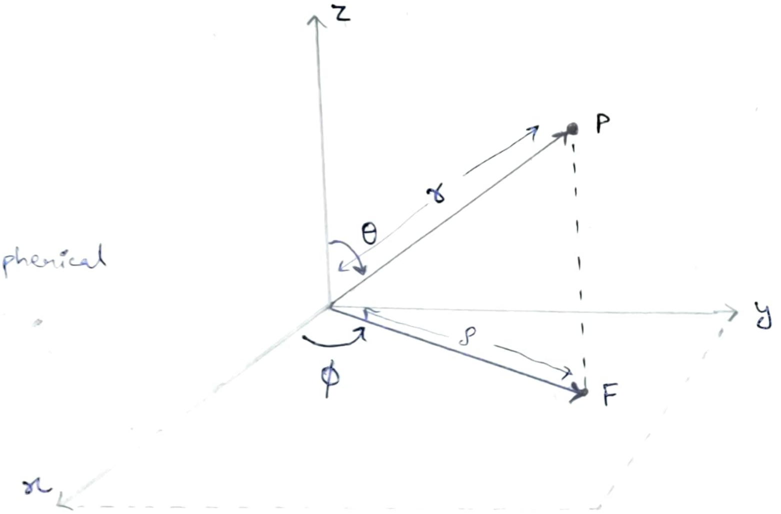
8) $\boxed{\begin{array}{l} \rho \rightarrow \text{variable} \\ \phi \rightarrow \text{variable} \\ z \rightarrow \text{variable} \end{array}}$ $\Rightarrow$ Solid Cylinder

SPHERICAL COORDINATE S/S



\# $O(0, \theta, \phi)$



$\boxed{\rho \neq r}$

of cylindrical — of spherical

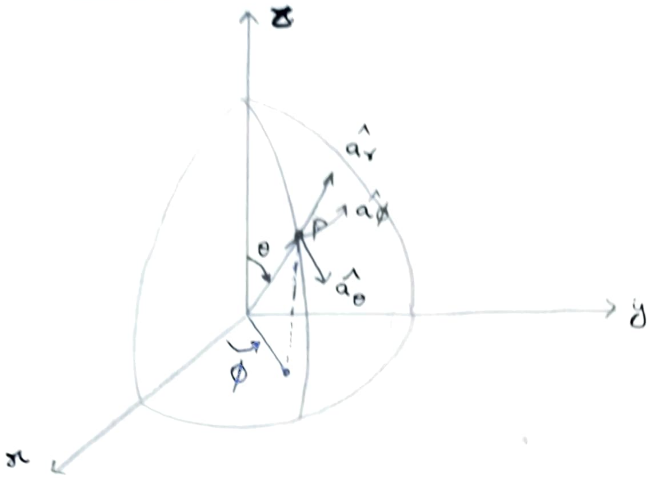
$r \rightarrow$ radius of imaginary sphere passing through point

$$\boxed{0 \leq r < \infty}$$

$\theta \rightarrow$ angle measured from +ve z-axis towards vector $\vec{OP}$. From TOP to BOTTOM $\rightarrow$ $\boxed{0 \leq \theta \leq \pi}$

$\phi \rightarrow$ same as cylindrical system $\rightarrow$ $\boxed{0 \leq \phi \leq 2\pi}$

Basis Vector or Unit Vector

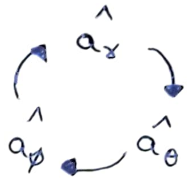


$$\boxed{\hat{a}_r \perp \hat{a}_\theta \perp \hat{a}_\phi}$$

Dot Product & Cross Product

→ Dot product :- diff → 0

same → 1

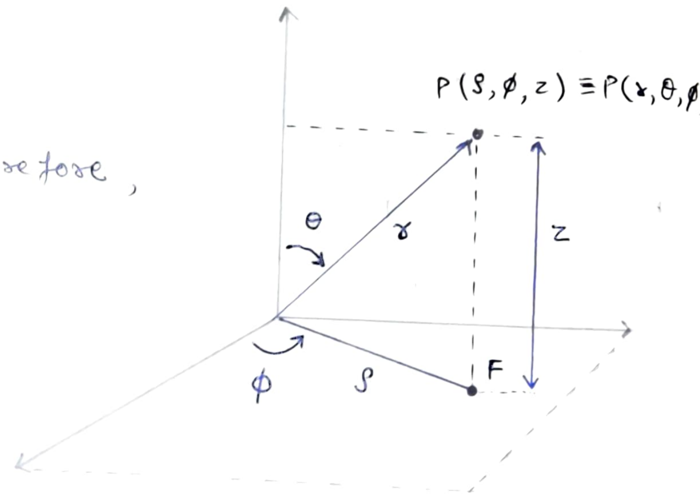


→ Cross product :- same → 0

diff → 3rd by following RHL

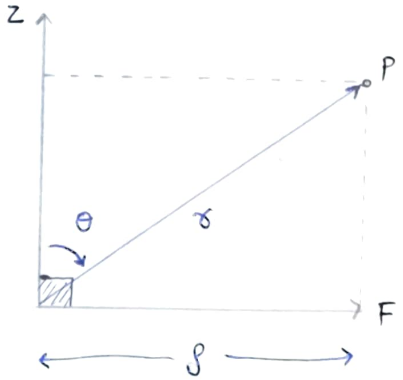
Conversion (Spherical ⟷ Cylindrical)

1) Point Conversion :



- z-axis ⊥ xy-plane. Therefore,

  z-axis ⊥r to $\vec{OF}$.



$\therefore$
$$\boxed{\begin{aligned} \rho &= r \sin\theta \\ z &= r \cos\theta \\ \phi &= \phi \end{aligned}}$$

$$\rho = r\sin\theta$$
$$\phi = \phi$$
$$z = r\cos\theta$$

Spheri $\longleftrightarrow$ cylind

$$r = \sqrt{\rho^2 + z^2}$$
$$\theta = \tan^{-1} \rho/z$$
$$\phi = \phi$$

cylind $\updownarrow$ spherical

2) Vector Conversion

$$\vec{A}_{cylind} = A_\rho \hat{a_\rho} + A_\phi \hat{a_\phi} + A_z \hat{a_z} \quad ; \quad \vec{A}_{sphere} = A_r \hat{a_r} + A_\theta \hat{a_\theta} + A_\phi \hat{a_\phi}$$

$$\rho = r\sin\theta$$
$$\phi = \phi$$
$$z = r\cos\theta$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

Transpose $\downarrow$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

3) Unit Vector Conversion

↳ Same as above matrix system.

DOT PRODUCT OF DIFF. C.S UNIT VECTOR

$$\rho = r\sin\theta$$
$$\phi = \phi$$
$$z = r\cos\theta$$

	$\hat{a_r}$	$\hat{a_\theta}$	$\hat{a_\phi}$
$\hat{a_\rho}$	$\sin\theta$	$\cos\theta$	0
$\hat{a_\phi}$	0	0	1
$\hat{a_z}$	$\cos\theta$	$-\sin\theta$	0

Q:- $\int_0^{\pi/2} (\hat{a}_\rho \cdot \hat{a}_r)\, d\theta = \int_0^{\pi/2} \sin\theta \, d\theta = \Big| -\cos\theta \Big|_0^{\pi/2} = \boxed{1}$

Spherical $\longleftrightarrow$ Cartesian

1) Point Conversion

$$x = \rho \cos\phi = r \sin\theta \cos\phi$$
$$y = \rho \sin\phi = r \sin\theta \sin\phi$$
$$z = z = r\cos\theta$$

Spherical to Cartesian

$$r = \sqrt{\rho^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$$
$$\theta = \tan^{-1}\frac{\rho}{z} = \tan^{-1}\left(\frac{\sqrt{x^2+y^2}}{z}\right)$$
$$\phi = \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

Cartesian to Spherical

2) Vector Conversion

$\{|A| = 1\}$

$$x = r\sin\theta\cos\phi$$
$$y = r\sin\theta\sin\phi$$
$$z = r\cos\theta$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = [\text{Transpose}] \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$